The Rope Programme: Formula Compendium

A single-place reference for the load-bearing relations of the rope corpus, organised by sector. Every entry carries its registry provenance and its status. Compiled at v3.17.0 (577 registered claims, 539 code-backed). The registry (claims.yaml) is the authority; where this document and the registry disagree, the registry wins.

How to read this document

STATUS grades, as used throughout the corpus: Derived — proved from stated premises, all of which are registered. Modeled — computed within a model whose premises are stated but not all derived. EFT-constrained — fixed by effective-theory matching. Conjecture — asserted with its conditionality on the face. Open — registered as unresolved. Failed-and-kept — refuted, retained on display.

Symbols. T (or T0) strand tension [N]; mu line density [kg/m]; Sigma stiffness / tension energy density [J/m^3]; a mesh spacing [m]; w strand width; kappa the locking modulus [J/m^2] (NOT the surface gravity kappa of the gravity sector, nor ELEC-021's Coulomb coefficient — a registered name collision, rename queued); K the coarse-grained XY stiffness [J]; J the per-link locking energy [J]; l_q the winding's source length; g = l_q/a; d_c strand thickness; c speed of light.

Conditionality. Everything vacuum-facing (Sigma_vac, the mesh pair (a, T0), and every number derived from them) is conditional on FND-037 at the FND-040 floors. Absolute scale is irreducible by the corpus's own theorem (FND-MATTER-005): one calibration has been spent (the electron mass), and it is spent once.

1. FOUNDATIONS — the medium's primitives

The framework starts from strands with tension, a mesh spacing, and a locking interaction at crossings. Everything else is built from these.

```
J = T^2 / kappa                          locking energy per endpoint (exact XY form)
```

FND-001, Derived. Endpoint force balance. [Under the PRED-003-ETA enslavement this sharpens to J = T a/2.]

```
K = J / a   (scalar);  K = 2J/a          coarse-grained stiffness (director form)
```

FND-002, Derived. Corrects an earlier 3J/a. Load-bearing for the coupling sector.

```
Pi = kappa a / T                         the single dimensionless coupling
```

FND-005, Derived. The parameter count: {T, kappa, a} carry exactly one dimensionless combination.

```
Sigma = 3 T0 / a^2   <=>   T0 = Sigma a^2 / 3
```

FND-017 / GRV-076, Derived invariance. The relation that lets the stiffness and the mesh pair be solved jointly.

```
E_vortex   = pi K ln(R/a_eff) + E_core,   a_eff = 0.18 a,  E_core = 5.448 K
```

FND-008/009 + benchmarks/micromech/defect_cores.py, Derived. The core cutoff is computed, not assumed.

```
E_pair     = 2 pi K ln(d/a)          vortex-antivortex (2D Coulomb, BKT)
```

FND-009, Derived.

```
S_eff      = (K/2) ∫ (grad theta)^2   leading-order effective action
```

EFT-001, EFT-constrained; Gamma-convergence of the discrete locking energy on vortex-free fields (FND-003).

```
u(x)      = T0 ( x/2 - x^2/8 + ... ),   x = (grad g)^2
```

FND-040, Derived. The arc-length expansion of an inextensible constant-tension strand — the NEGATIVE quartic is the medium's softening.

```
mu = T / c^2                          line energy equals tension
```

HBAR-001 / GRV-077, Derived identity; with c = sqrt(T/mu) it is a restatement, and is used as such.

2. ELECTROMAGNETISM — charge, fields, and the fine-structure chain

```
q  = Lk                               charge = topological linking number (integer)
```

EM-001 / GG-006, Derived. Charge is a conserved integer; the Coulomb field is rope-count flux geometry. The screw/torsion Goldstone carries CHARGE (winding); the separate collective transverse pair carries LIGHT (EM-RECON-025).

```
Maxwell    from Bianchi identity + Chern-Weil in d = 3
```

EM-003, Derived. The field equations are geometric identities of the bundle, not postulates.

```
Z0 = sqrt(mu0/eps0) = 376.73 ohm;     c^2 = 1/(mu0 eps0)
```

EM-002, Derived (structural constants). mu_0 = 1/(eps0 c^2) forced by the carrier wave equation via Faraday (EM-RECON-028).

```
Z_medium  = sqrt(T mu) = T / c        intrinsic rope impedance
```

OPT-006, Derived. Yields eps_med = 1/T — the medium's permittivity is the inverse tension (PRED-003-DICT).

```
alpha     = l_q^2 T / (4 pi hbar c)   the impedance reduction
```

PRED-003-DICT, Modeled. l_q is the winding's SOURCE length; equivalently hbar = T l_q^2/(4 pi alpha c).

```
kappa_lock = 2T/(eta a),  eta = 1  =>  kappa = 2T/a,  l_lock = a/2,  J = T a/2
```

PRED-003-LOCK / PRED-003-ETA, Modeled (conditional on P1 one-metric applicability, P2 single-symbol-T). eta = 1 by one-metric uniqueness: one wave operator cannot carry two stiffnesses.

```
alpha     = g^2 T a^2 / (4 pi hbar c),   g = l_q/a;   lambda = g^2/(4 pi)
```

PRED-003-ETA, Modeled. g is the corpus's single remaining mesoscopic unknown (FND-044).

```
g         = l_q/a = 43.0 · kappa_pack^(1/6)     [82.6 at the 5% floor, 108.0 continuum]
```

FND-041, Modeled. The sixth-root law: a ~ kappa^(-1/3) (FND-038/040) and l_q ~ T0^(-1/2) (R1).

```
kappa_0   = c / sqrt(eps0 Sigma)      circulation quantum normalization
```

EM-RECON-027/028, Modeled; the value is Sigma-normalization dependent (Sigma_eff-normalized after FND-038).

```
1/alpha   = 4 pi^3 D_E = 137.0605     (+178.8 ppm), D_E = 1.1051029 blind
```

ALPHA-DE-CHAIN, Modeled. A distinct route: no continuous dials, one discrete input, three named external gates.

```
sin^2 theta_W = 1/(3 sqrt 2) = 0.2357
```

EW-001, Modeled. From winding/Hopf geometry.

```
Delta n(par) : Delta n(perp) = 3 : 1,  sign NEGATIVE   (vacuum birefringence)
```

EM-RECON-016, Derived given the field mapping. A double qualitative discriminator against QED's 7:4 positive; PVLAS excluded the ATLAS identification.

3. LIGHT AND CLASSICAL OPTICS — the transverse channel

WHAT THE CARRIER IS, stated carefully, because the corpus corrected itself here. Light is the COLLECTIVE TRANSVERSE displacement mode of the mesh — the transverse Goldstone pair of broken translation symmetry, two polarization states (s_x, s_y), gapless at all crossing strengths (EM-RECON-025, Modeled). It is NOT the torsion/screw mode. GRV-020's Corollary 1 allocated "torsion dynamics = light" by topology, and that allocation OVER-REACHED: the single screw field carries ONE state against the photon's two and was killed on polarization by three computed identities — Malus washout, the Beth angular-momentum sign, and birefringence mode-count (EM-RECON-022/023, kill class 4). The tension was split into its own claim (GRV-102) and resolved in structure by Commission K. The screw mode retains its OTHER job: its winding is charge (pi_1(S^1) = Z, GG-006 — that half of Corollary 1 is untouched and load-bearing).

Live conditions on the carrier, on its face: the bending falsifier B <= T0 a^2/12, the mandatory three-pin fork, and the dispersive-spacing cost (a_disp <= 9.3e-28 m against a coverage scale of 6.0e-17 m). The optics results below ride on the transverse channel and were always written on it; nine of ten OPT claims are Derived, and they are unaffected by the carrier correction.

```
omega(k)  = c k                        non-dispersive propagation
```

EM-005, Derived. Phase velocity independent of k.

```
c         = sqrt(T/mu)                 wave speed from the medium
```

Standard string result on the corpus's primitives; verified pointwise in curved backgrounds to ~2% (GRV-048).

```
wavefront = envelope of secondary wavelets, advancing at c        (Huygens)
```

OPT-001, Derived.

```
I(theta)  ∝ sinc^2( pi a sin(theta) / lambda )      single slit
```

OPT-002, Derived.

```
I         ∝ 1 + cos( 2 pi Δ / lambda )              two-slit interference
```

EM-004, Derived, from linear superposition alone.

```
r = (Z1-Z2)/(Z1+Z2);   t = 2 Z1/(Z1+Z2)            Fresnel amplitudes
```

OPT-008, Derived, from the impedance mismatch.

```
theta_c   = arcsin(n2/n1)              total internal reflection
```

OPT-005, Derived.

```
tan theta_B = n2/n1                    Brewster angle
```

OPT-009, Derived.

```
omega_n   = n pi c / L;   f_1 = c/(2L);   omega_cutoff = pi c / a
```

OPT-003 / OPT-004, Derived. The mesh cutoff k <= 1/a is what makes cell-scale loops mode-free (FND-MATTER-053).

4. GRAVITY — induced metric and Einstein-Hilbert dynamics

Gravity is not postulated: it is the effective metric seen by transverse excitations, with its dynamics induced Sakharov-style from the weave band.

```
Zero-point theorem: a uniform network of ANY tension density is FLAT.
```

GRV-004, Derived. Rest tension does not gravitate; only deviations do. Structural — independent of Sigma's value.

```
mu u_tt = d_a( T_a d_a u )            the gapless transverse wave operator
```

GRV-029/055, Derived. Four coefficient functions (mu, T_x, T_y, T_z).

```
mu = B/alpha,  T_a = alpha B/b_a^2;   alpha B = sqrt(mu ∏ T)
   isotropic:  b = (T mu)^(1/4),   alpha = T^(3/4) mu^(-1/4)
```

GRV-029, Derived. An exact bijection on the positive cone: the photon sector is ONE-METRIC because there is no fifth function for a second metric to live in.

```
S_induced -> Einstein-Hilbert  (m-odd, IR-universal remainder; parameter-free
tensor pattern)
```

GRV-025, Derived. The absorption exam: measured tensor ratios match the EH pattern.

```
G         = c^3 a^2 / (16 pi zeta hbar),   zeta = 1.208
```

GRV-075/095, Modeled/Derived-conditional. The Sakharov route; a selected at the Planck class (a = 0.80 l_P after the GRV-101 correction).

```
1.751"  deflection, 43.0"/century Mercury, Shapiro delay, Nordtvedt —
unconditional within the corpus
```

GRV-029 consequence, Derived. gamma = 1 with the scalar screened at lattice range (GRV-028).

```
S = A/4  from  T_H = kappa/2pi  with  dM = T dS
```

GRV-039, Derived (numerically verified). Horizon thermodynamics from the ledger mechanism.

```
T_inf     = (beta K h / m*) kappa               the Hawking law-form
```

GRV-087, Modeled. sigma cancels exactly — the third cancellation.

```
A*        = (beta/0.23) Sigma a^3 h / (chi c);   n_q = A*/hbar = 1.1e-
4..4.6e-4
```

GRV-092, Modeled. The snap action is depth- and mass-blind: the horizon secretes a universal action constant. Sub-quantum — the whisper is classical radiation at leading order.

```
hbar      = T0 · [l_q^2/(4 pi alpha)] / c   and   A* = T0 · [(3 beta/(0.23
chi)) a h] / c
```

GRV-093, Modeled. ONE FORM — tension × area / c — across two sectors; the hbar gap is the ratio of two areas.

```
omega_ring = 0.23 kappa;   L = f (dM/dt) c^2 T_grey,  f <= 1e-2 (data-
bounded)
```

GRV-040 (demoted to T2 by GRV-049) / GRV-052, Modeled.

5. THE VACUUM-STIFFNESS TOWER — the v3.17.0 arc

All entries in this section are conditional on FND-037 (Conjecture) at the FND-040 floors.

```
Sigma_eff = 3.61–3.70e35 J/m^3        pinned by measurement
```

FND-030 / ELEC-052/081, Modeled. The historical 5.10e35 registration is demoted: one relation at a dead strand count.

```
Sigma_vac = kappa_pack · Sigma_eff,   kappa_pack >= 50 (5% CS) / >= 250 (continuum)
```

FND-037/040, Conjecture-grade floor. No ceiling (FND-042: the 1–100 window is HALF-WINDOW — lower edge physics, upper edge grammar).

```
a = (3 K_me / Sigma_vac)^(1/3),   T0 = K_me / a,   K_me = T0 a = m_e c^2/L = 2.6065e-14 J
```

FND-038/041, Modeled. The m_e-pinned solve; the single spent calibration, re-used never re-fit. (a, T0) = (1.63e-17 m, 1599 J/m) / (9.53e-18 m, 2734 J/m).

```
delta_D - delta_f = -(eps_f/2)(C_D/C_f - 1),   eps_f = 1/kappa_pack
```

FND-040, Derived from registered premises (inextensibility + the global tension multiplier). The sign is NEGATIVE: geometric softening.

```
kappa_pack = (C_D/C_f - 1) / ( 2 |delta_D - delta_f| )        THE PIN
```

FND-046, Modeled. Forward: delta_A - delta_F = -1.25% (kappa=50) / -0.25% (kappa=250). Current lattice bound (Bali 2000, <=5%) gives kappa_pack >= 12.5 (FND-047).

```
kappa_pack = ( (l_q/a) / 43.0 )^6                             the ratio inversion
```

FND-042, Modeled. Any external determination of l_q/a measures the vacuum packing at ×6 sensitivity.

```
N = C^2 eta = 2 g^2 = 3698 · kappa_pack^(1/3)                 the area-selection composite
```

FND-043/044, Modeled. The L1 question, the vacuum packing, the amplitude selection and lambda are ONE unknown: what sets g.

6. THE QUANTUM BOUNDARY — what is derived, and what is not

The Schrödinger equation is ADOPTED, not derived; the Born rule is OPEN as a three-part decomposition. What follows is what the corpus does derive.

```
CHSH <= 2 sqrt 2                      the Tsirelson bound, DERIVED
```

QB-019/020, Derived. Mechanism-unity (one response object per setting per wing) caps CHSH at exactly 2√2; Gram/TLM verified.

```
P(detect) = cos^2(theta/2) = (1 + a·n)/2      the detection law from strand holonomy
```

QB-024/025, Derived. The quaternion scalar part IS cos(theta/2): the belt trick as the origin of the amplitude.

```
CHSH_local-counting <= 2;  severed-strand wall E = -1/3
```

QB-006/013/023, Derived. The counting reading is theorem-capped; -1/3 is the signature of severed-strand averaging.

```
S         = pi T A^2 / (2c);   A_hbar = sqrt( 2 hbar c / (pi T) )
```

HBAR-001, Modeled. L cancels; hbar fixes exactly one number — the amplitude. A = 2.6348 ·l_q (ELEC-083 identity).

```
rho       = A / l_q = 1/sqrt(2 pi^2 alpha) = 2.6348   =>   1/alpha = 2 pi^2 rho^2
```

ELEC-083, Modeled. Under the shared-origin hypothesis alpha reduces to ONE pure geometric ratio; the hypothesis is carried, not adopted.

7. NUCLEAR STRUCTURE — composition, binding, and the droplet

```
M(A,Z)   = knot count − rope-mode-overlap binding
```

NUC-001, Derived-structure/Modeled-numeric: masses reproduced to <0.1% given measured binding energies.

```
atomic masses PREDICTED, C-12 to U-238: 0.00–0.51% (post-NUC-018 correction)
```

NUC-005 as amended by NUC-018, Modeled. a_V = 17.77 with the corrected spacing (2.026 fm) and surface ratio (1.34).

```
S/V bond ratio = (d/4)(36 pi rho)^(1/3) · ... ;  coordination number z CANCELS exactly
```

NUC-015, Derived. Every close-packed lattice gives 1.33–1.34 against an empirical 1.16 — the gap is registered, not hidden.

```
n_cross   = rho z d / 4                bonds cut per unit area, orientation-averaged
```

NUC-015, Derived.

```
R         = (3A / 4 pi rho)^(1/3)      droplet radius at saturation density
```

NUC-022, Modeled.

```
E_F term:  <E> = (3/5) E_F;  symmetry energy from N≠Z label counting
```

NUC-010/019/022, Modeled. The N=Z preference emerges from unlike-label bonding.

8. MATTER, ATOMS, AND CHEMISTRY

```
shell capacities = 2 n^2   (n^2 rope modes × two spin states)
```

CHEM-STRUCT-001, Derived (counting).

```
R         ~ a sqrt( N / 4 pi )        atomic pattern radius from rope count
```

FND-MATTER-002, Modeled — a consistency WINDOW, not a prediction (two free inputs, one target). Numerological fixes explicitly rejected.

```
N fixed by the interpenetrability coverage threshold (a geometric number, not a parameter)
```

FND-MATTER-004, Modeled.

```
lambda    = pi (r/a)^2 = 2.776e-6     zero-point suppression, dilute-perturbation Casimir
```

FND-MATTER-056, Modeled — a QUANTIFIED NULL: 4.16× from target, with the O(1) rescue pre-refused. lambda REMAINS OPEN.

```
IRREDUCIBILITY: the absolute mesh scale a is NOT derivable from the postulates.
```

FND-MATTER-005, Derived (a theorem about the framework). One calibration is spent; the corpus honours this rather than quietly contradicting it.

9. THERMODYNAMICS AND STATISTICAL MECHANICS

```
Z         = ∏ sqrt( 2 pi T / (K lambda_k) )      Gaussian partition function
```

THM-002, Derived; free energy, entropy and energy follow.

```
C         = 1/2 per mode               2D Dulong-Petit
```

THM-003, Derived.

```
T_BKT     = pi K / 2;   eta(T_BKT) = 1/4;   jump = 2 T_BKT / pi
```

THM-004/005, Derived. Standard BKT phenomenology on the corpus's own stiffness.

10. THE FALSIFIABLE RELATIONS (what could kill the programme)

```
d ln alpha / d ln G = fixed scale-free ratio  (−1, −2, or −3 by l_q composition)
```

PRED-003, Modeled — the corpus's sole T1. A measured nonzero ratio IDENTIFIES the drift channel and, through the FND-042 inversion, reads kappa_pack (FND-043).

```
|Gdot/G| > 1.6e-18 /yr at 3 sigma   REFUTES  (decision window 2027–2030, J1713)
```

PRED-003-CONF/META/J1713, Modeled.

```
delta_A - delta_F = -1.25% / -0.25%   long-distance Casimir-scaling violation, SU(3)
```

FND-046/047, Modeled. Kill conditions: quartic-DOMINANT violation kills; POSITIVE deviation kills the derived sign; eps_f > 5% kills the floor.

```
sigma_k < k sigma_1                   bundles bind (k unit windings attract)
```

FND-048, Modeled. The qualitative bundle prediction; matches every dataset under both candidate laws. The (N,k) magnitude is the named next frontier.

```
Delta n < 0 with 3:1 anisotropy       vacuum birefringence (PVLAS-class)
```

EM-RECON-016, Derived given the mapping; magnitude Sigma-suppressed.

11. THE OPEN NUMBERS — what the corpus does not have

An honest compendium ends with the blanks. Four Conjectures, six Open claims, thirty-four kept failures stand in the registry; the formula-level gaps are these:

```
g = l_q/a  — the mesoscopic source length. UNEXPLAINED.
```

FND-044. Five former questions collapse to it. Its best-shaped mechanism (defect-log energy budget) is EXCLUDED structurally: FND-045, Failed-and-kept, demand 2.2–3.2 m_e c^2 blind.

```
lambda — the zero-point suppression. OPEN.
```

FND-MATTER-050/055/056. Reframed: not "what weights the term" but "what suppresses mesh-scale vacuum energy by ~1e-5" — the matter sector's own hierarchy problem.

```
hbar — not derived. The question has a SHAPE: what selects the area l_q^2/(4 pi alpha)?
```

GRV-093 / FND-043. Narrowed to the locking normalization N; not closed.

```
m_e, and the absolute scale — irreducible by theorem, one calibration spent.
```

FND-MATTER-005 / FND-017.

```
Born rule, Schrödinger equation — adopted/open, not derived.
```

QB-007/008; see KNOWN_LIMITATIONS.md, which is the document to read before this one.

Compiled 2026-08-11 at v3.17.0. Provenance for every line is the claim ID given beneath it; run tools/verify_corpus.py and the cited benchmarks to check any of them. Failures are kept on display by design — see KNOWN_LIMITATIONS.md and the Failed-and-kept claims in claims.yaml.